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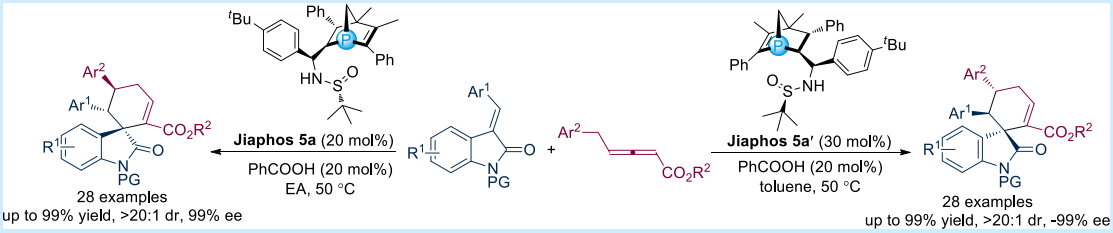
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Design of 1-Phosphanorbornene Derivatives as Chiral Organocatalysts for Enantioselective (4 + 2) Annulation Reactions of γ-Benzyl Allenoates

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ABSTRACT: Two novel diastereoisomeric P-chirogenic phosphine catalysts, i.e., JiaPhos, which can be easily derived from inexpensive and commercially available starting materials in five chemical operations (totally 4.16g scale), are introduced. To our delight, the JiaPhos catalysts display good performance in enantioselective (4 + 2) annulations involving 3-methylene-2-oxindoles and γ-benzyl allenoates, providing a wide range of 3,3′-spirocyclic oxindoles with good efficiency and enantioselectivity.

Phosphine-catalyzed reactions offer a very powerful tool for the synthesis of structurally diverse optically active organic building blocks.[1] As a result, the design of efficient chiral phosphine catalysts has attracted much attention.[2] In the past decades, many types of chiral phosphines have been developed by Lu,[3] Zhao,[4] Huang,[5] Fu,[6] Zhang,[7] and other organic chemists.[8] Although these phosphines showed excellent catalytic activities in asymmetric phosphine catalysis, efficient methodology for the design and synthesis of P-stereogenic phosphines remains scarce. An early example with the development of P-chirogenic catalysts was disclosed by the Zhang group in 1997.[9] Subsequently, Sasai's group designed a novel P-chirogenic catalyst with an *o*-hydroxyalkyl chelating arm, which exhibited good performance in the aza-Morita−Baylis−Hillman (aza-MBH) reaction.[10] In 2014, Kwon and co-workers developed a novel family of P-stereogenic [2.2.1] bicyclic chiral phosphines, applying them in the asymmetric installation of pyrrolines.[11] Recently, Zhang's group reported a novel class of cyclic phosphine-derived bifunctional phosphine catalysts (Le-Phos), which showed high enantioselectivity in γ-addition reactions of N-centered nucleophiles to allenoates (Scheme 1, A).[12] As a part of our program, we set our goal to *de novo* design novel P-stereogenic catalysts derived from the 1-phosphanorbornenes, in which three P−C bonds of the P-chirogenic center are absolutely shackled. In addition, the two alkyl groups link to the phosphorus atom that shows better nucleophilic activity than 1-phosphanorbornadienes,[13] and combining with a H-bonding site and multiple tunable side chains, the *de novo* design of P-chirogenic catalysts (i.e., Jiaphos) may extend the applicative scope of asymmetric phosphine catalysis to a greater extent (Scheme 1, B).

In 2003, Mathey and co-workers reported a Diels−Alder reaction of 2*H*-phospholes with *trans*-cinnamaldehyde.[14] The cycloaddition primarily took place at the C=O bond, and only minor C=C bond adducts could be obtained. The experimental results demonstrated that the carbonyl group of aldehydes could compete with activated alkenes in (4 + 2) cycloadditions. To our delight, when we applied *trans*-cinnamaldehyde dimethyl acetal as a reactive substrate, the cycloaddition completely took place at the C=C bond, affording the desired compounds efficiently. The phosphine was then protected by further vulcanization. The mixture of *exo*/*endo*-**1** was obtained in a total of 61% yield, which could be straightforwardly separated by column chromatography, giving the *exo*-**1** in 52% yield. Next, deprotection of *exo*-**1** in CH_2Cl_2 at 0 °C produced the desired compound **2** (94% yield). Two sets of diastereomeric chiral sulfinyl imines **3** could be obtained in a total of 80% yield by the condensation reaction of compound **2** with (R_S)-*tert*-butanesulfinamide. The stereoselective addition of ArMgBr to compound **3** followed by reduction with Raney Ni then afforded two sets of

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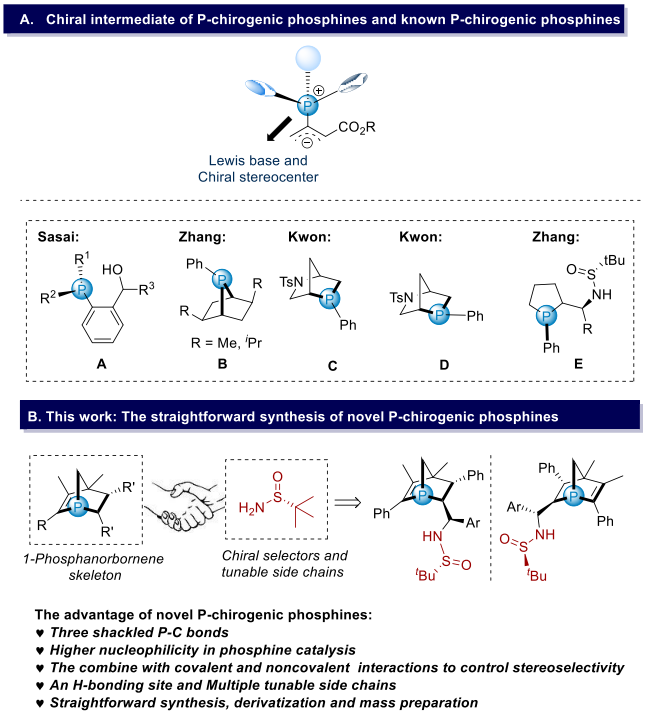


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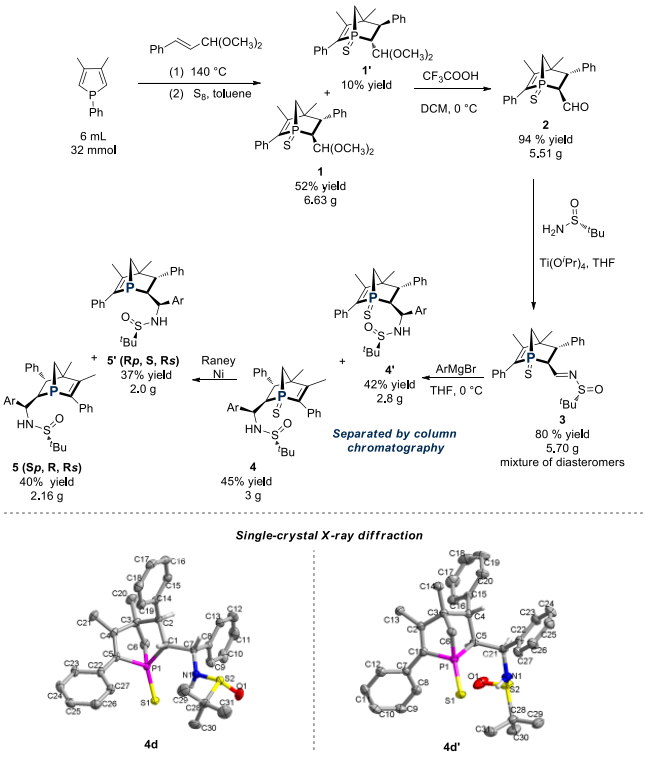
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Scheme 1. Design of Novel P-Chirogenic Catalysts

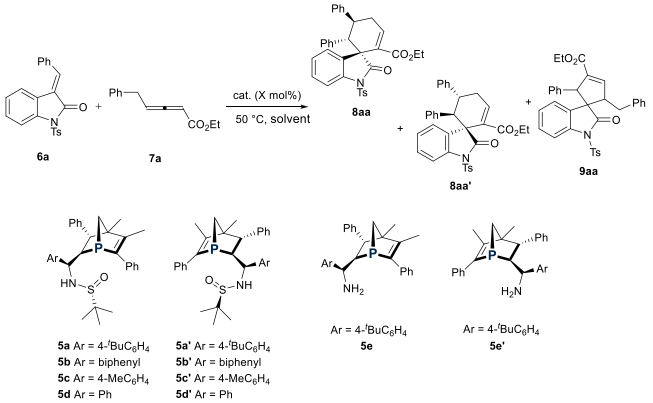


Scheme 2. Synthesis of P-Chirogenic Phosphine JiaPhos



diastereomeric JiaPhos derivatives **5** and **5′** with high yields, which could be straightforwardly separated by column chromatography. To test the practicality of this new methodology, a gram-scale synthesis was carried out. To our delight, the 32.0 mmol scale reaction produced 2.16 g of (R_p,S,R_S)-**5** and 2.0 g of (S_p,R,R_S)-**5′** in a total of 15.6% and 14.5% yield by five steps, thus indicating that this transformation was easy to scale-up to gram scale without loss of efficiency and selectivity

Table 1. Optimization of Phosphine-Catalyzed (4 + 2) Annulation of γ-Benzyl Allenoate 7a[a]



entry	cat. (%)	solvent	yield(%)[b]	8aa:9aa[c]	ee[d]
1	**5a** (20)	toluene	93	23:1	99
2	**5a** (20)	EA	99	25:1	97
3[e]	**5a** (20)	EA	95	32:1	99
4[e]	**5b** (20)	EA	98	>20:1	98
5[e]	**5c** (20)	EA	95	>20:1	85
6[e]	**5d** (20)	EA	93	>20:1	97
7[e]	**5e** (20)	EA	86	20:1	86
8	**5a′** (20)	toluene	51	15:1	−98
9	**5a′** (25)	toluene	65	14:1	−99
10	**5a′** (30)	toluene	72	13:1	−98
11[f]	**5a′** (30)	toluene	82	18:1	−98
12[f]	**5a′** (30)	EA	24	9:1	−99
13[f]	**5a′** (30)	CH_3CN	25	3:1	−89
14[f]	**5a′** (30)	$CHCl_3$	10	5:1	−98
15[f]	**5a′** (30)	THF	77	7:1	−99
16[f]	**5a′** (30)	dioxane	74	8:1	−99
17[e,f]	**5a′** (30)	toluene	95	>20:1	−98
18[e,f]	**5b′** (30)	toluene	82	>20:1	−99
19[e,f]	**5c′** (30)	toluene	93	>20:1	−99
20[e,f]	**5d′** (30)	toluene	93	>20:1	−99
21[e,f]	**5e′** (30)	toluene	62	>20:1	−96

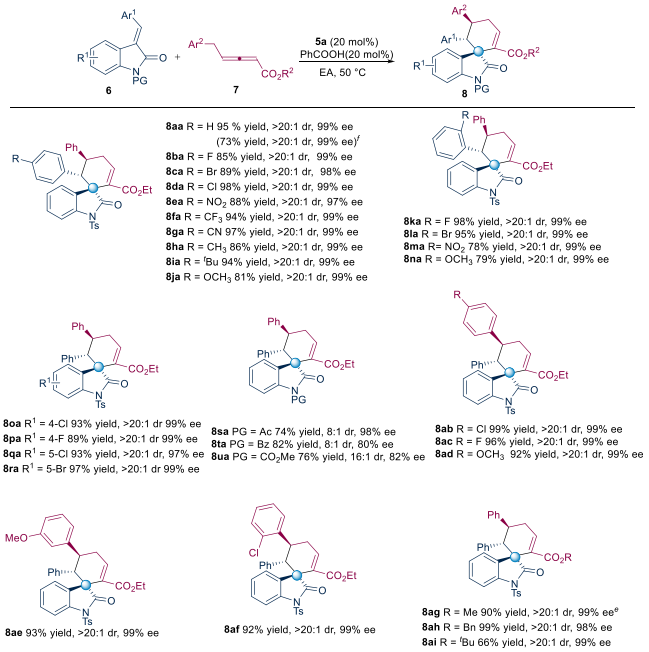
[a]Reactions were performed with **6a** (0.1 mmol), **7a** (0.2 mmol), and the catalyst in the solvent specified (1.0 mL) at 50 °C. EA = ethyl acetate. Reaction time: 12 h. [b]Isolated yield. [c]Determined by ^{1}H NMR. [d]Determined by HPLC analysis on a chiral stationary phase. [e]PhCOOH (0.02 mmol) was used. [f]Reactions were performed with **6a** (0.1 mmol) and **7a** (0.3 mmol).

(Scheme 2). The absolute configurations of the two sets of JiaPhos catalysts were established by single-crystal X-ray diffraction analysis (**4d** and **4d′**).

With the P-chirogenic phosphine JiaPhos in hand, the performance of Jia-Phos catalysts was focused on the asymmetric catalysis. In 2013, Marinetti and Huang's group reported a novel phosphine-catalyzed (4 + 2) annulation of γ-benzyl allenoates, respectively.[15] However, the development of enantioselective variants of these transformations has met with very limited success. For Huang's (4 + 2) reaction, Lu's group used NUSIOC-Phos as a catalyst, and only 39% ee was obtained.[16] For Marinetti's (4 + 2) reaction, only 50% yield and 66% ee were obtained with (*S*,*S*)-BDPP. Thus, we chose γ-benzyl allenoates and 3-methylene-2-oxindoles as potential synthetic precursors to test its use in enantioselective phosphine catalysis.[17] The initial reaction of ethyl γ-benzyl allenoate **7a** with (*E*)-3-benzylidene-1-tosylindolin-2-one **6a** in

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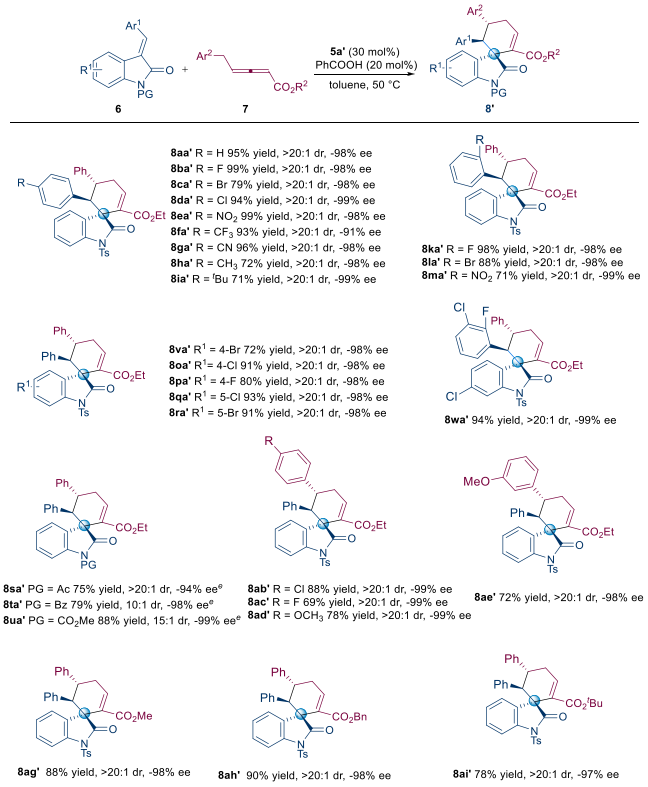
Scheme 3. Enantioselective Allene−Olefin (4 + 2) Annulations Catalyzed by 5a[a,b,c,d]



[a]Reactions were performed with **1** (0.1 mmol), **2** (0.2 mmol), benzoic acid (0.02 mmol), and the catalyst **5a** in ethyl acetate (1.0 mL) at 50 °C at 12 h. [b]Isolated yield. [c]dr was determined by ^{1}H NMR. [d]Determined by HPLC analysis. [e]Reaction temperature: 80 °C. [f]**6** (1 mmol) and **7** (2 mmol) were used.

the presence of catalyst **5a** afforded the product **8aa** in 93% yield with 99% ee and minor (3 + 2) product **9aa** (**8aa**:**9aa** = 23:1) (Table 1, entry 1). When ethyl acetate is used as the solvent, we found that a similar result could be produced at 50 °C in the presence of catalyst **5a** (Table 1, entry 2). From a green and sustainable chemistry point of view, we finally chose ethyl acetate as the solvent. The addition of benzoic acid led to improvements in the chemoselectivity (Table 1, entry 3). Next, different JiaPhos **5a**−**5e** catalysts were examined further, and the catalysts **5a** proved to be the best choice (Table 1, entries 3−7), affording the desired product in 95% yield with 99% ee (Table 1, entry 3). Two sets of chiral phosphines **5** and **5′**, albeit with diastereoisomeric property, possess opposite stereochemistry at their P-chiral centers at which the covalent bonds to the allenoates were formed, and we thus anticipated that they might give respective 3,3′-spirocyclic oxindole antipodes. To our delight, the catalyst **5a′** could also provide the opposite 3,3′-spirocyclic oxindole enantiomer in 51% yield with −98% ee (Table 1, entry 8). The increase of catalyst loading could improve the yield (Table 1, entries 8−10). To further enhance the results, the screening of solvents suggested that toluene was identified as the solvent of choice (Table 1, entries 11−16). The addition of benzoic acid still led to an improvement in both chemical yield and chemoselectivity (Table 1 entries 11 vs 17). In almost all of the catalysts **5a′**−**5e′** examined, the excellent enantioselectivity could be obtained, albeit with moderate to excellent yields (Table 1 entries 17−21). Remarkably, JiaPhos **5a′** promoted the formation of 3,3′-spirocyclic oxindole in 95% yield as well as −98% ee under the optimized reaction conditions (Table 1, entry 17).

Scheme 4. Enantioselective Allene−Olefin (4 + 2) Annulations Catalyzed by 5a′[a,b,c,d]



[a]Reactions were performed with **6** (0.1 mmol), **7** (0.3 mmol), benzoic acid (0.02 mmol), and the catalyst **5a′** in toluene (1.0 mL) at 50 °C at 12 h. [b]Isolated yield. [c]dr was determined by ^{1}H NMR. [d]Determined by HPLC analysis. [e]Reaction solvent: THF.

With the established optimal reaction conditions in hand, we first explored the reaction scope in the presence of catalyst **5a**, and the results are summarized in Scheme 3. The aryl group in **6** was well tolerated, regardless of the electronic nature and substitution pattern of the aryl moieties of the methyleneindolinones, producing the 3,3′-spirocyclic oxindoles in high yields and nearly perfect enantioselectivities (Scheme 3, **8aa**−**8na**). The enantioselective cyclization reactions of **6o**−**6r**, which contain 4-Cl, 4-F, 5-Cl, and 5-Br, on the indole ring, respectively, were all converted to the corresponding products in excellent yields with excellent enantioselectivities (Scheme 3, **8oa**−**8ra**). The N-protection of indole rings (N-Ac, N-Bz, and N-CO_2Me) could be also utilized. In almost all of the examples examined, 3,3′-spirocyclic oxindoles **8** were produced in good chemical yields and good to excellent enantioselectivities (Scheme 3, **8sa**−**8ua**). Various ethyl γ-benzyl allenoates bearing *o*-, *m*-, and *p*-substituents in the phenyl rings were suitable substrates, and the desired products **8** were isolated in 92−99% yields with 99% ee (Scheme 3, **8aa**−**8af**). To further investigate the substrate scope, the ester groups of γ-benzyl allenoates were tested, and the yield slightly diminished when we applied bulky steric *tert*-butyl γ-benzyl allenoates, retaining excellent enantioselectivities (Scheme 3, **8ag**−**8ai**).

We then examined the reaction scope with regard to catalyst **5a′**, and the results are summarized in Scheme 4. The reaction was well-tolerated with aryls of methyleneindolinones containing different *para*- and *ortho*-substituents, from electron-donating to electron-withdrawing groups (Scheme 4, **8aa′**−

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8ma′). Not surprisingly, when the methyleneindolinones bearing 4-Br, 4-Cl, 4-F, 5-Cl, and 5-Br on the indole ring were employed, respectively, the reaction could produce their corresponding 3,3′-spirocyclic oxindoles in high yields and excellent ee (Scheme 4, **8va′**, **8oa′**−**8ra′**, and **8wa′**). When we used the methyleneindolinones containing N-Ac, N-Bz, and N-CO_2Me protection, the substrates were all found to be suitable, although slightly lower diastereoselectivities were obtained (Scheme 4, **8sa′**−**8ua′**). Moreover, the phenyl ring at the γ-benzylic position of allenoates could be replaced by other aryls. Allenoates bearing an aryl moiety with *meta* and *para* substituents were all found to be suitable in this transformation (Scheme 4, **8aa′**−**8ae′**). Finally, we applied different ester groups of γ-benzyl allenoates as substrates, and the corresponding products could also be obtained with high yield and excellent ee (Scheme 4, **8ag′**−**8ai′**). The alkyl-substituted methyleneindolinones (alkyl instead of Ar^1) were also used as substrates to react with γ-benzyl-substituted allenoates, and no product was formed.

In conclusion, we have developed a highly concise and efficient strategy for the construction of a new class of rigid P-chirogenic phosphine catalysts, which are highly effective catalysts for the asymmetric phosphine-catalyzed annulations of γ-benzyl allenoates. To the best of our knowledge, this is the first systematic report on the phosphine-catalyzed asymmetric (4 + 2) annulation of γ-benzyl allenoates. Both enantiomers of the products could be obtained in good yields and with excellent enantioselectivity through transformations that were catalyzed by two diastereomers of Jia-Phos. Further application of these catalysts to other transformations is underway.

■ ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge on the ACS Publications Web site at: Crystallographic data for **4d**, **4d′**, **8la**, **8ai′**(CIF) Experimental procedures, X-ray diffraction data for **4d** (2033734), **4d′** (2033735), **8la** (2033733), **8ai′**(2033730) and (PDF) are available free of charge via the Internet at . The Supporting Information is available free of charge at https://pubs.acs.org/doi/10.1021/acs.orglett.1c00833.

> Experimental procedures, X-ray diffraction data for **4d**, **4d′**, **8la**, and **8ai′**, and spectroscopic data for all new compounds including ^{1}H, ^{31}P, and ^{13}C NMR spectra (PDF)

Accession Codes

CCDC 2033730, 2033733−2033735, and 2049497 contain the supplementary crystallographic data for this paper. These data can be obtained free of charge via www.ccdc.cam.ac.uk/data_request/cif, or by emailing data_request@ccdc.cam.ac.uk, or by contacting The Cambridge Crystallographic Data Centre, 12 Union Road, Cambridge CB2 1EZ, UK; fax: +44 1223 336033.

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Author Contributions

All authors have given approval to the final version of the manuscript.

Notes

The authors declare no competing financial interest.

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